

Profession: A profession is a paid job that needs **special training** and **formal qualifications**.

Professional: A professional is a person who **belongs to a profession** or **earns money** by doing a specific skilled job. Professionals follow **education standards, training**, and have the **skills and knowledge** needed to do their role properly.

Professional Responsibilities

In **IT, Computer Science, and Software Engineering**, professionals are responsible for: Network setup and management, Software development, installation, and maintenance, Managing the technology life cycle of a company (upgrading, replacing hardware/software)

According to Engineering Council, an Engineer must know: Technical knowledge + good decision-making. Government laws related to the job (safety, health, environment). Basics of management and industrial relations. Information about trade unions. Responsibilities to the profession, society, community, and environment. Basic knowledge of taxes (income tax, sales tax, property tax)

Professionalism: Professionalism is not just what you do — it is who you are. It is a way of thinking, behaving, and presenting yourself that shows commitment to excellence. It includes: Positive attitude, Being coachable (ready to learn, Taking responsibility for your actions. Very important for juniors and startups to show professionalism early.

Characteristics (Traits) of a Profession: Four Main Traits: Activities that require special skills, Society-centric motivation, Personal standards of excellence, Giving back to society

Traits Explained

- 1. Special Skills** Every profession needs its own skills. Example: Computer programmers have different skills from electrical engineers.
- 2. Society-Centric Motivation:** How your profession helps or impacts society. Example: A software engineer develops attendance systems, payroll systems, e-commerce sites, etc.
- 3. Personal Standards of Excellence:** Some engineers grow fast (e.g., become IT managers) because they keep improving; others stay in the same position for years.
- 4. Giving Back to Society:** Do your job honestly and contribute at least **2.5%–5%** of your effort to society and humanity (helping others, contributing positively).

Ethics: Ethics is about human behavior — how a person acts in society or a company. Examples of ethical behavior: Saying “Sorry” when you make a mistake. Saying “Thanks” or appreciating others when they do something good

Law: Law is a set of rules enforced by an authority. Example: A company has rules about office timing, deadlines, discipline. Law protects both employer and employee. If employees break laws, they are punished; if they follow rules, they should be rewarded (bonus, appreciation).

Law: Enforced by authority. If you break rules, you can be penalized or face action

Ethics: Personal behavior. If you act poorly, people form a bad impression, but you may not be punished

Morality (part of ethics) examples: “How are you?” “You look great in this dress.” “You were absent yesterday, everything okay?”

Properties / Characteristics of a Professional

- **Property #1 — Seriousness:** Be responsible. Complete tasks on time. Follow requirements. Don’t wait for reminders or warnings to finish your work
- **Property #2 — Wanting to Do Better:** Improve your performance every day. Don’t stay at the same level — grow with each project
- **Property #3 — Dealing With the Unexpected:** Problems and changes will happen. Professionals adapt and handle changes. This helps build a successful career
- **Property #4 — Communication Skills** Two types: **Verbal, Written**
- Good communication must be: **Clear** (easy to understand), **Concise** (short and to the point), **Confident** (sure about what you’re saying)
- **Property #5 — Enthusiasm:** Attitude matters. People with energy and a positive attitude become leaders
- **Property #6 — Helpfulness:** Success requires teamwork. Always ready to help. Give suggestions. Encourage others with compliments or appreciation
- **Property #7 — Taking Initiative:** Don’t be afraid to take the first step. People who take initiative move ahead faster
- **Property #8 — Keeping Cool Under Pressure:** Stay calm during stressful situations. Business and jobs always have pressure. Stress harms performance; staying calm improves it
- **Property #9 — Staying Focused:** Focus on one task at a time. Avoid multitasking. Even during difficult times, stay steady and consistent

- **Property #10 — Be a Trend Setter, Not a Trend Follower.** Analyze situations. Try new ideas and solutions. Leaders take new paths and create trends

Applying Professionalism in Real Life

Case #1: You are an IT Director. Your team wants training and certifications.

Professional response (legal, moral, ethical): Appreciate their interest in learning. Check company policies and budget. Arrange training if possible. If not possible immediately, plan a future roadmap. Be honest and supportive

Case #2: Your employees want to work for an NGO during company time.

Professional response: Check company rules and project deadlines. Company time should be used for company work. Offer them to work for the NGO after office hours. If company supports charity, you may allow limited involvement legally and ethically

Case #3: You are an Android developer, and management asks for 2 extra hours daily.

Professional response: Check your employment contract and labor laws. Extra hours should be voluntary and compensated fairly. Politely discuss workload or request overtime pay/compensation. Maintain ethics, avoid arguments, communicate respectfully

Week 2 : Professional Ethics & Code of Ethics

Law: Laws are **rules created by a company or society** that tell people what they **must do** or what they **must not do**. Laws help maintain order and control behavior.

Moral Values: Moral values are **principles of right and wrong**. They guide a person's decisions and behavior. Example: Honesty, kindness, respect.

Ethics: Ethics means **socially acceptable behavior**. Example: Respect for employees, respect for customers.

- **Code of Ethics:** A Code of Ethics explains the **responsibilities of professionals** to act ethically. Applies to employees, companies, and society. Organizations create and distribute these policies and expect workers to follow them.
- **Goals of Code of Ethics** Helps people make good decisions. Includes: Intellectual Property, Privacy, Confidentiality, Professional Quality, Fairness or Discrimination, Liability (Responsibility), Software Risks, Conflicts of Interest, Unauthorized Access to Computer Systems

Detailed Goals of Code of Ethics

1. **Intellectual Property:** Related to creations of the mind. Examples: Software, algorithms, flowcharts, programs, DFDs, research papers, books, patents.
2. **Privacy:** Do not interfere in other people's matters. Example: If managers are in a meeting, don't enter or listen to their discussion.
3. **Confidentiality:** Keep official matters **secret**, do not talk about them publicly. Example 1: After a meeting, keep the termination list confidential. Example 2: System administrators must keep user IDs and passwords secret.
4. **Professional Quality:** Your work should be of **good quality**. People should not complain about it.
5. **Fairness or Discrimination:** Treat everyone equally without bias. Example: A manager should promote the employee who deserves it, not the one who is a personal favorite.
6. **Liability (Responsibility):** Take full responsibility for the tasks assigned to you. Be an **asset**, not a liability for your company.
7. **Software Risks:** Every software project has risks in: Analysis, Design, Development, Implementation
Examples: Requirements changing (analysis stage) Not enough resources (implementation stage)
8. **Conflicts of Interest:** People can have different opinions. Use **logic and reasoning** to solve conflicts.
9. **Unauthorized Access:** Accessing systems without permission is **unethical and a crime**. Every employee should access only their own computer or system using their user ID and password.

IEEE: IEEE = **Institute of Electrical and Electronics Engineers**. Largest technical professional organization. Promotes development in Electrical, Electronics, Computer Science, and IT. Works for the benefit of humanity and improvement of the profession.

IEEE Code of Ethics IEEE ethics relate to: Actions, Products, Responsibilities, Peers Self

1. **IEEE Code of Ethics – Actions: Public:** Software engineers must act in the public interest. **Clients & Employers:** They must work in the best interest of clients/employers while keeping public interest first.
2. **IEEE Code of Ethics – Products:** Software engineers must ensure that their software and updates meet highest professional standards.
3. **IEEE Code of Ethics – Responsibilities: Judgment:** Maintain honesty and independence in decisions. **Management:** Managers must promote ethical practices in software development and maintenance.

4. **IEEE Code of Ethics – Peers: Profession:** Increase honesty and reputation of the profession. **Colleagues:** Be honest and supportive to colleagues.
5. **IEEE Code of Ethics – Self:** Engage in **lifelong learning**. Promote ethical practices for improving the profession.

Principles of IEEE Code of Ethics: Act in public interest, Act in interest of clients/employers, Produce quality products, Maintain independent judgment, Manage ethically, Protect honesty of profession, Support colleagues, Pursue lifelong learning

ACM (Association for Computing Machinery): World’s largest educational and scientific computing society. Brings together computing educators, researchers, and professionals. Goal: Share resources, discuss issues, and solve challenges in computing.

- **ACM Code of Ethics – General Moral Values**

ACM members must: Contribute to society and well-being, Avoid harming others, Be honest and trustworthy, Be fair; avoid discrimination, Respect copyrights, agreements, and property rights, Respect intellectual property, Respect others’ privacy, Maintain confidentiality

- **ACM Code of Ethics – Professional Responsibilities**

A computing professional must: Try to achieve **highest quality** in work, Improve processes and products, Know and follow professional laws, Give complete evaluation of system risks, Honor contracts and responsibilities, Access systems only when authorized, Obtain and maintain professional competence

- **ACM Code of Ethics – Organizational Leadership**

A leader must: Explain social responsibilities to employees. Manage people and resources to create quality systems. Ensure computing resources are used properly. Ensure developed systems meet user requirements. Make sure manpower is properly used. Arrange training for employees working on computer systems

Ethical Decision-Making Cases

Case 1 Situation: Alisha, a network manager, is asked to install a cheaper but less secure system. She believes the data is too sensitive for a low-security product.

Questions & Answers

Q1: Which code applies? **A1:** Security and Unauthorized Access

Q2: What should Alisha do? **A2:** (i) Convince management to buy a secure system. (ii) If they refuse, she should resign

Case 2 Situation: Ahmed, a programmer, used programs from 3 teammates but didn’t acknowledge them. He also added code from his friend Rashid but didn’t inform management.

Q1: Which code applies? **A1:** Intellectual Property

Q2: Did Ahmed follow ethics? **A2:** No, because: (i) He did not appreciate team members (ii) He violated copyright by using Rashid’s code secretly

Case 3 Situation: A software house in Canada delivered a tax software with errors. Clients paid wrong taxes and were penalized.

Q1: Which code applies? **A1:** Violated ACM code – did not honor contracts and responsibilities

Q2: Software house mistakes: **A2:** (i) They did not test software properly. (ii) They made false agreements with clients

Case 4 Situation: Management tells a software manager to hire only white people and prefer females over males.

Q1: Which code applies? **A1:** Fairness and Discrimination

Q2: What should you do? **A2:** (i) Explain disadvantages of discrimination (ii) If management insists, refuse to take part in hiring

Week 3: Software uses, Misuses, Software Development risks

Computer Program: A computer program is a set of instructions written in a programming language for a computer to execute. It is one part of software. Programs are ordered and sequential instructions. Example languages: C++, Java, Python.

Computer Software: Software is a **set of instructions, data, or programs** that help computers run and perform tasks.

Uses of Software: Produces results very fast. Produces accurate results without errors. Helps prevent fraud. Helps store and retrieve data from databases. Increases employee productivity Examples: Microsoft Word, Excel, PowerPoint, Human Resource Management, Payroll systems, Inventory control, Financial accounting, Production control, Library systems

Misuses of Software: Using software without permission or pirated software violates the copyright act. Software is intellectual property of the creator or company. To use someone else’s software legally, a license agreement must be signed.

Software Development Risks: Software development has **internal and external risks**. Identifying risks is **important for quality** and achieving project goals. Developers must **understand risks** to respond effectively.

List of Software Development Risks: Code issues, Aggressive deadlines, Inaccurate estimations, Low productivity, Budget issues, Poor risk management, Inadequate project management, Change in project scope, Stakeholder issues, User response, Project team members leaving, External risks

Details of Software Development Risks

1. **Code Issues:** Poor quality code, inconsistent style, missing documentation. **Errors:** syntax, logical, runtime. **Solutions:** Manual and automated testing. Fix errors when found. Create coding standards. Use object-oriented techniques
2. **Aggressive Deadlines:** Tight deadlines may cause teams to fail. **Solution:** Make a realistic project plan. Set achievable deadlines
3. **Inaccurate Estimations:** Wrong estimates for cost, deadlines, or outcomes can cause failure. **Solution:** Discuss risks with clients/end-users. Clarify chances of achieving estimates. Use MS Project to schedule tasks
4. **Low Productivity:** Delays or employee stress can reduce productivity. **Solutions:** Well-designed project plan to reduce stress. Clear communication about project details. Strong leadership to motivate the team
5. **Budget Issues:** Costs can increase if project scope changes. **Solution:** Monitor project spending. Adjust project plan and budget as scope changes
6. **Poor Risk Management:** Ignoring risks can cause failure. **Solutions:** Identify potential risks. Calculate probability of each risk. Create plans to avoid risks. Monitor risks carefully
7. **Inadequate Project Management:** Poor management can lead to unclear goals and deadlines. **Solutions:** Hire a good project manager. Develop a detailed project plan. Use project management software. Communicate clearly with stakeholders
8. **Change in Project Scope** Large or sudden changes affect deadlines and costs. **Solution:** Divide project into manageable segments. Manage and review each segment regularly
9. **Stakeholder Issues:** Stakeholders can influence project success. **Problems:** Lack of interest. High expectations **Solutions:** Communicate frequently and clearly. Keep stakeholders engaged
10. **How Users Respond:** Project success depends on user adoption. **Solutions:** Test software thoroughly. Conduct surveys. Organize focus groups to gather feedback

SMART Goals in Software Development: Goals should be **SMART:** Specific, Measurable, Attainable, Realistic, Timely

Week 4 - Cyber Crime & Intellectual Property

CYBER CRIME: Any criminal activity in which a computer, mobile phone, tablet, or even a landline is used as the main tool.

Forms of Cyber Crime

1. **Fraud:** A wrongful or criminal act done to gain money or personal benefit. *Examples:* A bank employee transfers staff salaries into his own account and escapes. A cashier collects customer cash but does not enter it into the POS system.
2. **Embezzlement:** When someone responsible for handling money accepts payments but does not record the transaction and keeps the money. *Example:* Cashier takes money but does not enter it in the POS machine.
3. **Software/Information Sabotage:** Intentional attempt to damage or corrupt software or data. *Examples:* Infecting a website. Sending viruses targeting MBR or EXE files
4. **Information (Data) Theft:** Illegal transfer or storage of confidential, personal, or financial information. *Examples:* Stealing someone's ID, password, or keys. Unauthorized access to personal accounts
5. **Hacking and Cracking:** Unauthorized access to computer systems or networks.
6. **Digital Forgery (Digital Tampering):** Manipulating digital documents or images for financial, political, or social gain.

Hacking: Attempt to access a computer or network without permission. Hackers may aim to gather information for illegal purposes. Hackers are intelligent programmers who use their skills negatively. Hacking is done using software/programs.

Cracking: Unauthorized access to steal, corrupt, or view data illegally. Crackers are *not* programmers — they use techniques like guessing PINs or passwords, breaking user IDs, etc.

Difference Between Hacking & Cracking

Hacking	Cracking
Access systems; may or may not cause harm	Access systems <i>to steal, corrupt, or view data</i>
Hackers are programmers	Crackers are not programmers
May be curiosity-based	Always illegal and harmful

Digital Forgery: Manipulating digital documents, images, or files. Used for financial, social, or political advantage. Gives misleading information about a product or service.

Security Measures: Track and record user access with servers. Assign unique ID and password to every user. Passwords must include letters, numbers, and special characters. Users should change passwords regularly. Only authorized people should be allowed in sensitive areas like server rooms. Maintain log files with details such as date, time-in/out, name, and purpose of entry.

INTELLECTUAL PROPERTY (IP): Creations of the human mind — creative or intelligent — such as art, designs, music, literature, symbols, programs, images, etc.

Examples of Intellectual Property: Painting (Art), Building design, Novel (literary work), New song (music), Computer program, algorithm, flowchart, Research papers, Symbols (e.g., red signal = STOP), Images and scenery

Types of Intellectual Property:

1. **Copyright:** Protects original creative works. Prevents others from copying or producing similar items. Applies to: Books, Journals, magazines, bulletins, Songs and music, Dramas, theatre productions, Movies, TV programs, talk shows. Pakistan Copyright Act: Copyright Ordinance, 1962 (amended in 2000)
2. **Trademark:** Unique symbol or sign that identifies a product or company. Example: McDonald's logo. Represents the identity of the product. Illegally copying a trademark is punishable by law.
3. **Patent:** Government authorization that gives exclusive rights to make, use, or sell a product for a fixed time. Only the approved party can produce or sell the item. Examples: *Jam-e-Shirin* – Qarshi, *Rooh Afza* – Hamdard Pakistan, *Student Biryani*, *Mazaidar Haleem*
4. **Trade Secret:** Secret formulas or processes that give a company an advantage. If kept private, the protection can last forever. Example: Coca-Cola formula — never patented, kept secret for decades.

Week 5 Cyber Law and Intellectual Property Law

Intellectual Property Rights (IPR)

- Intellectual property and the rights related to it are often **the most valuable assets** for any company, organization, or institution.
- Things that must be protected include: All **information saved on computers or electronic devices** All **paperwork and documents** like: Computer programs, User manuals, Information on websites
- Unauthorized use of intellectual property must be **stopped through law**, and damages should be recovered.
- Since technology keeps upgrading, the law also keeps changing.
- To handle this, **GATT (General Agreement on Tariffs and Trade)** was created by 23 founding members.
- GATT:
 - Covers **intellectual property rights**
 - Is the only **multilateral and legally binding** agreement
 - Is a **legal agreement between countries and companies**
 - Covers **90% of world trade**, although some countries still haven't joined

Does GATT Cover Intellectual Property?

- GATT's purpose: Promote international trade by **removing trade barriers** like tariffs and quotas.
- Article XX(d) of GATT 1947 / 1994 mentions **intellectual property rights**.
- In Pakistan, the following laws support and follow GATT: Copyright Ordinance 1962, Patent Ordinance 2000, Trade Marks Ordinance 2001

Confidential Information (Covered Under GATT)

- Confidential information = information that is **not public** and **should not be shown to the public**.
- Examples: Personal information (e.g., email password). Trade secrets. Sensitive government information. Work handled by computer scientists like: Algorithms, Computer programs. Any information that must be protected from unauthorized access. **Non-confidential information** can be used by anyone.

Plagiarism: Taking someone else's work or ideas and presenting them as your own. The following actions are counted as plagiarism: Claiming someone's work as your own. Copying words or ideas without credit or reference. Using someone's work without citation

Cybercrime: Any crime involving a **computer or computer network**. A computer may: Be used to commit the crime Or be the target of the crime . Cybercrime can harm: Hardware, Software, Data, Finances

Types of Cybercrime include: Fraud, Embezzlement, Software/Information Sabotage, Data Theft, Hacking & Cracking, Digital Forgery

Cyber Law: The area of law that deals with **all types of cybercrimes** carried out through the internet. It aims to prevent unauthorized access involving: Computers & mobile phones, Privacy, Emails & communications, Intellectual property, Websites, Software & hardware, Databases & storage devices

- Cyber laws differ from country to country. Punishments may include **finances and imprisonment**.

Cyber Law in Pakistan: Pakistan established NR3C (National Response Centre for Cybercrime) under FIA. NR3C investigates cybercrimes, tracks criminals, and works to prevent internet misuse.

Punishments for Cybercrime: For spreading false information about someone: Up to 3 years in prison, OR up to PKR 1 million fine, OR both. **If someone** intentionally and publicly posts information that is: False. Hurting someone's reputation or privacy → This also counts as a punishable cybercrime.

Cybercrime Through Social Media: Rafay Baloch, a Pakistani security researcher, is known for ethical hacking and Android OS research. Social media platforms like Facebook, LinkedIn, Twitter, WhatsApp, Instagram, TikTok, Twitch, Ramble, etc., are commonly used for cybercrimes.

Examples of cybercrimes on social media: Phishing attacks (stealing passwords, financial details). Identity theft. Fake news about individuals, groups, government, or private organizations. Online scams. Cyberbullying (mental and emotional harm)

Cyber Laws in the World: Different countries have different cyber laws, for example: **USA** → Federal Trade Commission Act (FTCA). **Ireland** → Electronic Commerce Act. **UK, USA, Australia, New Zealand, Singapore** → Electronic Transactions Act. **Hong Kong** → Electronic Transactions Ordinance. **India** → Information Technology Act. **Bangladesh** → Information Communication Technology Act (Draft)

12. Cyber Laws in Pakistan

Pakistan has various cyber laws that deal not only with internet crimes but also with all areas related to computers and networks. Two well-known cyber laws are: **Electronic Transaction Ordinance 2002, Electronic / Cyber Crime Bill 2007**

Week 6- Software House & Its Goals

Software House: A software house is a company that mainly focuses on developing and releasing software products. It creates software for businesses or general consumers, depending on what their clients need. Their main job is to develop, maintain, and distribute software solutions.

Goals of a Software House: Software houses work for their clients using different technologies and platforms such as: PC and laptop software, Mobile applications, Online applications, Web development, Data science algorithms, APIs, etc. Clients may include large, medium, and small organizations, and each client's needs depend on their business nature and company size.

Principal Goals of a Software Development Team A software development team aims to: Build high-quality software, Meet client requirements, Use efficient development processes, Improve and update products continuously, Deliver reliable and scalable software

Types of Software Houses

1. **Product-Based Software Houses:** These companies create their own products and sell them to customers. They keep improving and upgrading their products to stay competitive. They provide ready-made (out-of-the-box) solutions. Examples: Microsoft, Google, HP, Facebook, Adobe, Cisco.
2. **Service-Based Software Houses:** They **build custom software** based on what the client needs. They provide **consulting services** to help organizations choose the right software solutions. They may also provide dedicated development teams to work directly with clients. **Examples of service providers:** TCS, FedEx, InfoSys, Wipro, etc. **Examples of custom software they build:** Inventory control system, HR system, Payroll system

How to Register a Software House (Pakistan)

1. **Choose a suitable company name:** Name should represent your company and its goals.
2. **Register with SECP:** SECP = Security and Exchange Commission of Pakistan. This legally registers the company.
3. **Register with PSEB** PSEB = Pakistan Software Export Board. Required for IT and software export companies.
4. After successful registration, you will receive: **Incorporation Certificate, Company NTN Certificate** from FBR (Federal Board of Revenue)

Week 7- Key Features of a Software House

Software House Hierarchy: A software house has a **structured hierarchy** (organogram) that shows how people work together, who is responsible for what, and how communication flows from top to bottom.

Key Features of a Software House Modern software houses have several important features: Focus on quality, Agile methodology, Full-stack development capabilities, Strong communication and collaboration, Flexibility, Proper hierarchical / organizational structure, Centralized and decentralized operations, Appraisals for employees, Monitoring financial performance, Software cost tracking, Software revenue tracking, Project costing, Project sale, Sales report generation, Long-term planning. All of these help a software house manage projects smoothly and run the business effectively.

Focus on Quality: A good software house always delivers high-quality code. Quality work builds trust and brings more customers. Since software houses compete globally, they must ensure fast, efficient, and easily modifiable code.

High-quality code means: Fast performance, Easy to update or modify

Agile Methodology: Agile is the most widely used modern development approach around the world. Older methods like Waterfall are considered outdated. Agile keeps developers closer to clients, resulting in better feedback and improved results.

Full-Stack Capabilities: A reliable software house handles the **entire development process**, including: Frontend, Backend, Databases, UI/UX, Deployment. Clients receive a **complete finished product** under one contract.

Communication and Collaboration: Regular and smooth communication is essential for project success. Strong communication ensures better teamwork and fewer misunderstandings. Software houses often assign a project manager who acts as a bridge between the client and development team, improving trust and engagement.

Flexibility: Software development requires new, innovative, and updated technologies. Teams must stay up-to-date with the latest tools. Updated versions of software, libraries, and tools should always be encouraged.

Hierarchical Structure (Organogram): A typical software house hierarchy looks like this:

Centralized Operations: Decisions made for the entire company at one central point. Example: The Chairman sets office timings for all employees.

Decentralized Operations: Each department makes its own decisions independently. Example: Different attendance systems for departments with part-time and full-time employees.

Appraisals: Employees' achievements and contributions are recorded properly. Staff understands what is expected of them. Helps in giving **promotions, bonuses, and rewards**. Training and career development plans are also regularly reviewed.

Monitoring Financial Performance: Every month, the software house audits its financial status. If **revenue > expenditures**, the company is doing well. If **expenditures > revenue**, corrective actions are taken. Financial performance includes:

- **Cost:** Money spent on project development such as manpower, equipment, servers, etc.
- **Revenue:** Total income earned.
- **Overheads:** Additional expenses like maintenance, utilities, salaries of non-technical staff, etc.

Software Cost: Total amount spent to **create and deliver software** to the customer. Includes: Hosting fees, Server/cloud charges, Website development and maintenance, Software licenses, Developer team salaries

Software Revenue: Total money **earned** for the software delivered to a customer. Revenue amount is usually written and agreed upon in a **contract** before development starts.

Project Costing: The cost and income of each project are calculated every month. Profit is calculated as: $\text{Profit} = \text{Revenue} - \text{Cost}$

Software Sale: A software sale is a financial transaction between the buyer and the software house. The software is sold under a contract or agreement where both parties agree on a fixed price.

Sales Report: A document that shows all sales activities of the software house. Sales reports help identify new market opportunities and improve marketing strategies.

Long-Term Planning: Strategic planning for the future. Successful software houses plan for future market needs to increase revenue. Long-term planning includes: Setting long-term goals. Identifying future problems, hurdles, and risks in the development process

